

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0403

Kapstone 138 kV
December 25, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0403. The Interconnection Customer proposes a 87.1 MW wind facility interconnecting at the Kapstone 138 kV substation in the Virginia area. The request was received on November 17, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 240 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,292,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Louisa Generation Facility 345/138 kV TX, the Firebantz-Mount Pleasant 138 KV loads to 106.4% of its 236 MVA rating. Project DFAX: 6.1%.

The Nuclear Fuels-Alvin 138 KV is overloaded to 103.6% of its 201 MVA emergency rating for the loss of the Daytn-Chatom 345 KV. The project contributes 7.7% DFAX (6.7 MW) to this overload.

Purdy-Chatom 138 KV L: 102.7% loading (356 MVA) following the outage of the Louisa Generation Facility-Grimes 138 KV Line; DFAX = 6.2%.

For the loss of Fluvanna P.S-Louisa Generation Facility 345 KV, the Louisa Generation Facility 345/138kV Transformer loads to 102.0% of its 1,004 MVA rating. Project DFAX: 34.1%.

For the loss of Kapstone Substation-Innovative Solar 31 138 KV, the Kapstone-Grimes 138 KV loads to 101.2% of its 896 MVA rating. Project DFAX: 74.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Firebantz-Mount Pleasant 138 KV	Reconductor line	\$1,898,000	5% (\$94,000)
Alvin Substation-Nuclear Fuels 138 KV	Reconductor line	\$3,611,000	8% (\$300,000)
Purdy-Chatom 138 KV Line	Reconductor line	\$1,681,000	4% (\$73,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Louisa Generation Facility 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$11,500,000	5% (\$527,000)
Kapstone-Grimes 138 KV	Reconductor line	\$3,810,000	8% (\$298,000)
	Total allocated to project		\$1,292,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.